

PROJECT TOPIC: ENDANGERED SPECIES OF INDIA (FLORA AND FAUNA)

Topic Description: Comprehensive Tabular Study of Ten Endangered Species of India (5 Plants and 5 Animals): IUCN Status, Habitat, Threats, and Conservation Interventions

India is recognized internationally as one of the world's 17 mega-biodiverse nations, harboring four recognized global biodiversity hotspots: the Himalayas, the Western Ghats, the Indo-Burma region, and the Nicobar Islands (Sundaland). However, rapid industrialization, extensive forest fragmentation, illegal poaching, over-harvesting of medicinal herbs, and global climate change have driven hundreds of indigenous species to the brink of biological extinction.

This project documents a systematic scientific investigation into **Ten Endangered Species of India**, comprising **Five Endangered Plant Species** and **Five Endangered Animal Species**, classified in accordance with the International Union for Conservation of Nature (IUCN) Red List and Schedules I and II of the Wildlife (Protection) Act, 1972.

By compiling detailed botanical and zoological records—spanning scientific taxonomy, geographical distributions, habitat specificities, primary anthropogenic threats, and dedicated recovery programs—this study highlights the critical importance of species-specific recovery action plans and legal habitat protection across India's fragile biomes.

2. Importance of the topic (विषयाचे महत्त्व)

Documenting endangered species is vital for understanding ecological stability, loss of genetic diversity, and statutory conservation laws:

Key Reasons Why Endangered Species Matter :-

- **Preventing Irreversible Trophic Collapse:** Extinction is forever. Every species occupies a unique, irreplaceable ecological niche. The loss of a keystone herbivore or top predator triggers cascading trophic collapses that destabilize entire ecosystems.
- **Loss of Unexplored Pharmacological Wealth:** Endangered plants like *Rauvolfia serpentina* (Sarpagandha) and *Taxus wallichiana* contain irreplaceable chemical alkaloids used to formulate life-saving anti-cancer and hypertension medications that cannot be synthesized from scratch.
- **Statutory & Legal Enforcement:** Wildlife protection laws (Wildlife Protection Act, 1972) derive their enforcement power from certified endangered species lists. Understanding Schedule I protections enables citizens to assist in combating illegal wildlife trade.
- **Flagships for Landscape Preservation:** Saving charismatic endangered species like the Great Indian Bustard or Bengal Tiger automatically protects millions of hectares of fragile grasslands and river catchments.

Conserving endangered species is an ethical commitment to intergenerational justice and planetary stewardship.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The research project was formulated around structured biological, taxonomic, and conservation objectives:

Structured Project Objectives :-

- **Taxonomic & Threat Status Verification:** To identify, catalog, and verify the IUCN Red List status and Wildlife Protection Act schedule for 5 endangered plant and 5 endangered animal species.
- **Geographic & Habitat Distribution Mapping:** To document the specific biomes, states, and protected sanctuaries where remnant populations of these ten species survive.
- **Anthropogenic Threat Analysis:** To analyze the root causes of their decline, including habitat loss, commercial poaching, chemical bioaccumulation, invasive species, and overhead power-line collisions.
- **Project Recovery Evaluation:** To evaluate dedicated government recovery projects, such as Project Tiger, Project Great Indian Bustard, and captive botanical gene banks.
- **Student Action Plan Formulation:** To design practical student measures for combating illegal wildlife trafficking and participating in native plant conservation.

The scope encompasses IUCN databases, MoEFCC wildlife reports, and botanical survey monographs.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized secondary scientific research, taxonomic compilation, and threat assessment protocol was deployed:

Methodological Research Protocol :-

- **Phase 1 – IUCN & WPA Database Search:** Queried the official *IUCN Red List of Threatened Species* database (www.iucnredlist.org) and Schedule I of the amended Indian Wildlife (Protection) Act, 2022.
- **Phase 2 – Species Selection & Balance:** Selected 5 prominent flora species (trees, herbs, gymnosperms) and 5 fauna species (mammals, birds, reptiles, aquatic fauna) representing distinct Indian ecosystems.
- **Phase 3 – Tabular Data Compilation:** Extracted standardized indicators: Common Name, Scientific Name, Taxonomic Family, IUCN Status, Geographical Range, Major Threat Factors, and Special Features.
- **Phase 4 – Recovery Project Audit:** Researched government action plans, including ex-situ captive breeding centers, tissue culture propagation, and anti-poaching camera traps.
- **Phase 5 – Report Synthesis:** Structured findings according to the Standard XI Maharashtra State Board EVS format.

All biological nomenclature and statuses were verified against the Botanical Survey of India (BSI) and Zoological Survey of India (ZSI).

5. Observations (निरीक्षणे)

Systematic compilation of scientific records yielded a verified tabular documentation of ten endangered Indian species:

5.1 Tabular Record of 10 Endangered Species of India (5 Flora & 5 Fauna)

Species & Taxonomy	IUCN Status & WPA	Geographic Range & Habitat	Primary Threats & Conservation Features
Sarpagandha <i>Rauvolfia serpentina</i> (Apocynaceae)	Endangered (BSI Red List)	Western Ghats, sub-Himalayan moist deciduous forests	Over-harvested for reserpine alkaloid (blood pressure drug); wild populations decimated; tissue culture propagation underway.
Red Sanders <i>Pterocarpus santalinus</i> (Fabaceae)	Endangered (IUCN Red List)	Endemic to Seshachalam hills, Andhra Pradesh	Heavily trafficked for valuable heartwood; international smuggling syndicates; protected under CITES Appendix II and armed forest guards.
Himalayan Yew <i>Taxus wallichiana</i> (Taxaceae)	Endangered (IUCN Red List)	Temperate Himalayas (1,800–3,300m elevation)	Leaves and bark stripped for 'Taxol' (anti-cancer medicine); extremely slow natural regeneration; over-exploited across northern states.
Pitcher Plant <i>Nepenthes khasiana</i> (Nepenthaceae)	Endangered (Schedule I)	Endemic to Khasi & Garo hills, Meghalaya	Carnivorous plant surviving in nitrogen-poor soils; habitat destroyed by coal mining; botanical curiosity leads to illegal collection.
Ebony / Tendu <i>Diospyros celebica</i> (Ebenaceae)	Critically Endangered	Dense rainforest pockets of Western Ghats	Felled for jet-black heavy timber used in luxury musical instruments; slow-growing climax species with low seed viability.
Great Indian Bustard <i>Ardeotis nigriceps</i>	Critically Endangered (Schedule	Thar Desert (Rajasthan) & semi-arid grasslands of	Under 150 individuals remaining; low frontal vision causes fatal collisions with overhead high-voltage solar

6. Analysis of data (माहितीचे विश्लेषण)

Ecological analysis of the documented endangered species reveals common systemic drivers pushing species toward biological extinction:

Core Analytical Inferences :-

- **Commercial Black-Market Poaching & Smuggling:** For species like the Indian Pangolin and Red Sanders, their primary threat is not natural predation, but organized international wildlife crime. Superstitious beliefs regarding pangolin scales and demand for red sanders wood drive ruthless poaching despite Schedule I penalties.
- **The Grassland Conservation Blindspot (The GIB Crisis):** Indian conservation policy has historically prioritized dense forests (Project Tiger) while neglecting arid grasslands, dismissing them as 'wastelands'. The near-extinction of the Great Indian Bustard (under 150 birds left) occurred because its semi-arid grassland habitat was turned over to massive renewable energy projects and high-voltage power transmission grids.
- **Aquatic Habitat Fragmentation:** The Gharial and Ganges River Dolphin prove that rivers cannot be protected in disconnected fragments. Building barrages and sand-mining shallow nesting banks disrupts seasonal migration and destroys reproductive sandbars.
- **Pharmacological Over-Exploitation of Flora:** Medicinal plants like Sarpagandha and Himalayan Yew were harvested destructively by uprooting whole plants or stripping bark for pharmaceuticals, rather than practicing sustainable wild collection.

The analysis proves that saving endangered species requires strict habitat protection combined with global enforcement against trafficking.

7. Results and Conclusions (निष्कर्ष)

Conclusion: India's endangered species are precious, irreplaceable living legacies of millions of years of evolution. Saving them demands a national commitment: undergrounding high-voltage power lines in bustard habitats, protecting riverbeds from sand mining, enforcing anti-poaching laws, and establishing ex-situ gene banks.

Action Checklist for Protecting Endangered Species :-

- ❑ Never buy wildlife souvenirs, corals, ivory, snake leather, or medicines containing pangolin scales.
- ❑ Support campaigns advocating for underground power cables in Great Indian Bustard habitats.
- ❑ Purchase only certified farm-cultivated Ayurvedic herbs, rejecting unverified wild-harvested roots.
- ❑ Report online wildlife trafficking advertisements on social media to the Wildlife Crime Control Bureau (WCCB).
- ❑ Plant indigenous threatened tree saplings in campus botanical gardens to aid ex-situ conservation.

8. Bibliography / References (संदर्भ सूची)

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2. Botanical Survey of India (BSI). (2020). *Red Data Book of Indian Plants*. Kolkata.
3. Zoological Survey of India (ZSI). (2021). *Critically Endangered and Endangered Animals of India*. Kolkata.
4. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.